# REB, The Course
# Class 10 Learning Activity Solution

An ideal, 50 gal CSTR is fed a liquid solution containing 0.014 lbmol A $gal^{-1}$ and 0.020 lbmol B $gal^{-1}$ at a temperature of 70 °F. The heat capacity of the reacting liquid is constant and equal to 65 BTU $ft^{-3}$ $°R^{-1}$. A jacket surrounding the reactor is supplied with saturated steam at 220 °F. The steam condenses in the jacket, leaving as a saturated liquid. The overall heat transfer coefficient is 60 BTU $ft^{-2}$ $°R^{-1}$ $h^{-1}$, and the heat transfer area is 13 $ft^2$. Reactions (1) and (2) take place within the steady-state CSTR. The heat of reaction (1) is 45,000 BTU $lbmol^{-1}$, and the heat of reaction (2) is 39,500 BTU $lbmol^{-1}$. The rate expressions for reactions (1) and (2) are given in equations (3) and (4). The rate coefficients obey the Arrhenius expression. For reaction (1) the pre-exponential factor is $1.6 \times 10^{18}$ gal $lbmol^{-1}$ $h^{-1}$, and the activation energy is 46,000 BTU $lbmol^{-1}$. For reaction (2) the pre-exponential factor is $4.5 \times 10^{18}$ gal $lbmol^{-1}$ $h^{-1}$, and the activation energy is 54,000 BTU $lbmol^{-1}$. What volumetric feed rate will maximize the rate of production of reagent D? At that feed rate, what is the conversion of A and the reactor temperature?

$$A + B \rightarrow D + Z \quad (1)$$

$$D + B \rightarrow U + Z \quad (2)$$

$$r_1 = k_1 C_A C_B \quad (3)$$

$$r_2 = k_2 C_D C_B \quad (4)$$

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***Assignment Summary***

<u>Reactions</u>

$$A + B \rightarrow D + Z \quad (1)$$

$$D + B \rightarrow U + Z \quad (2)$$

Rate Expressions

$$r_1 = k_1 C_A C_B \tag{3}$$

$$r_2 = k_2 C_D C_B \tag{4}$$

Reactor: liquid-phase, steady-state, non-isothermal CSTR

Given Constants: $V$ = 50 gal, $C_{A,in}$ = 0.014 lbmol gal$^{-1}$, $C_{B,in}$ = 0.020 lbmol gal$^{-1}$, $T_{in}$ = 70 °F, $\breve{C}_p$ = 65 BTU ft$^{-3}$ °R$^{-1}$, $T_{ex}$ = 220 °F, $U$ = 60 BTU ft$^{-2}$ °R$^{-1}$ h$^{-1}$, $A$ = 13 ft$^2$, $\Delta H_1$ = 45,000 BTU lbmol$^{-1}$, $\Delta H_2$ = 39,500 BTU lbmol$^{-1}$, $k_{0,1}$ = 1.6 × 10$^{18}$ gal lbmol$^{-1}$ h$^{-1}$, $k_{0,2}$ = 4.5 × 10$^{18}$ gal lbmol$^{-1}$ h$^{-1}$, $E_1$ = 46,000 BTU lbmol$^{-1}$, $E_2$ = 54,000 BTU lbmol$^{-1}$.

Parameter: $\dot{V}_{in}$

Deliverables: $\dot{V}_{in,opt} = \underset{\dot{V}_{in}}{\arg\max}\left(\dot{n}_D\right), f_A\Big|_{\dot{V}_{in}=\dot{V}_{in,opt}}$, and $T\Big|_{\dot{V}_{in}=\dot{V}_{in,opt}}$

***CSTR Model Function***

CSTR Design Equations:

$$0 = \dot{n}_{A,in} - \dot{n}_A - r_1 V = \epsilon_1 \tag{5}$$

$$0 = \dot{n}_{B,in} - \dot{n}_B - \left(r_1 + r_2\right) V = \epsilon_2 \tag{6}$$

$$0 = -\dot{n}_D + \left(r_1 - r_2\right) V = \epsilon_3 \tag{7}$$

$$0 = -\dot{n}_U + r_2 V = \epsilon_4 \tag{8}$$

$$0 = -\dot{n}_Z + \left(r_1 + r_2\right) V = \epsilon_5 \tag{9}$$

$$0 = \dot{Q} - \dot{V}_{in}\breve{C}_p\left(T - T_{in}\right) - \left(r_1 \Delta H_1 + r_2 \Delta H_2\right) V = \epsilon_6 \tag{10}$$

CSTR Model Variables: $\dot{n}_A$, $\dot{n}_B$, $\dot{n}_D$, $\dot{n}_U$, $\dot{n}_Z$, and $T$

Additional Computable Unknowns: $\dot{n}_{A,in}$, $r_1$, $k_1$, $C_A$, $\dot{V}$, $C_B$, $r_2$, $k_2$, $C_D$, $\dot{n}_{B,in}$, and $\dot{Q}$

$$\dot{n}_{i,in} = \dot{V}_{in} C_{i,in} \quad i = A \text{ and } B \tag{11}$$

$$k_j = k_{0,j} \exp\left(\frac{-E_j}{RT}\right) \qquad j = 1 \text{ and } 2 \tag{12}$$

$$C_i = \frac{\dot{n}_i}{\dot{V}} \quad i = A, B, \text{ and } D \tag{13}$$

$$\dot{V} = \dot{V}_{in} \tag{14}$$

$$\dot{Q} = UA\left(T_{ex} - T\right) \tag{15}$$

Additional Uncomputable Unknown: $\dot{V}_{in}$

ATE Solver Requirements:

1. Initial guesses for the reactor model variables (provided as an argument to the CSTR model function)
2. $\dot{V}_{in}$ must be made available to the residuals function as a global variable.
3. Residuals function that
   a. receives a guess for the reactor model variables ($\dot{n}_A$, $\dot{n}_B$, $\dot{n}_D$, $\dot{n}_U$, $\dot{n}_Z$, and $T$)
   b. calculates the additional unknowns using equations (3), (4), (11) - (15).
   c. evaluates and returns the CSTR design equation residuals using equations (5) - (10).

## ***Deliverables Function***

1. Define a range of values for $\dot{V}_{in}$

$$\underline{\dot{V}}_{in} = \left[10, \cdots, 100\right] \text{ gal h}^{-1} \tag{16}$$

2. Define an initial guess for the CSTR model variables for the first value of $\dot{V}_{in}$

$$\dot{n}_{i,guess} = \dot{n}_{i,in} \qquad i = A, B, \text{ and } Z \tag{17}$$

$$T_{guess} = T_{in} + 5K \tag{18}$$

3. Loop through the values of $\dot{V}_{in}$ and

a. Call the CSTR model function, passing the initial guess as an argument, to solve the CSTR design equations
b. Save the corresponding outlet molar flow rate of reagent D
c. Set the initial guess equal to the current design equations solution

4. Identify the value of $\dot{V}_{in}$ where the outlet molar flow of D was maximized
5. Call the CSTR model function to solver the CSTR design equations for the inlet volumetric flow rate

$$\dot{V}_{in,opt} = \arg\max_{\dot{V}_{in}} \left(\underline{\dot{n}}_D\right) \tag{19}$$

6. Use the solution to calculate the corresponding conversion.

$$f_A\Big|_{\dot{V}_{in,opt}} = \frac{\dot{n}_{A,in} - \dot{n}_A\Big|_{\dot{V}_{in,opt}}}{\dot{n}_{A,in}} \tag{20}$$

### *Implementation of the Calculations*

The Python script, activity_10.py, and Matlab function, activity_10.m, that were written to perform the calculations accompany this solution.

### *Results and Discussion*

The calculations were performed as described above. When the molar flow rate of D was plotted vs. the inlet volumetric flow rate, a maximum was found in the original range of values of $\dot{V}_{in}$ around 45 gal $h^{-1}$. To improve accuracy, the calculations were repeated using a range from 35 to 50 gal $h^{-1}$. The results are shown in Figure 1, and the maximum molar flow of D was found to occur at a volumetric feed rate of 44.8 gal $h^{-1}$.

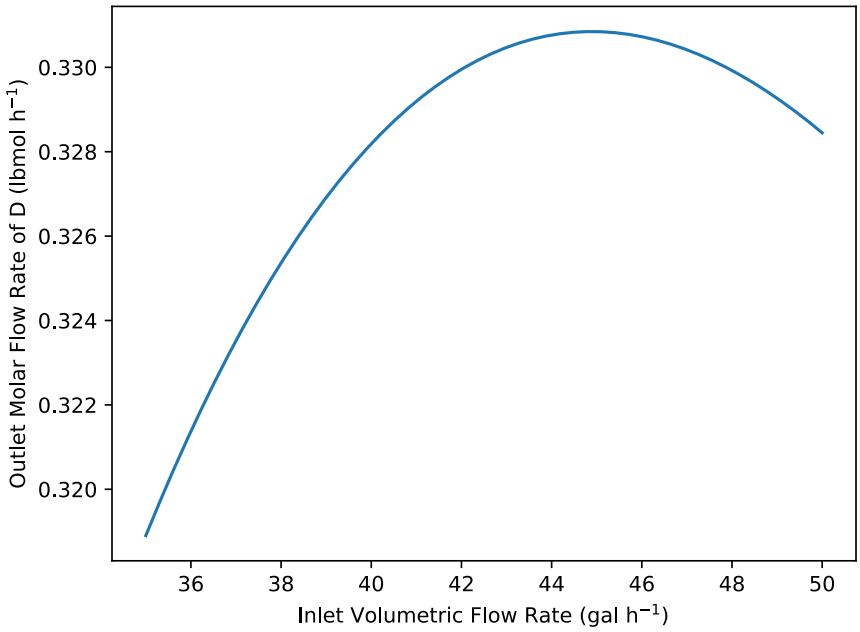



*Figure 1. The molar flow of D is maximized at an inlet volumetric flow rate of 44.8 gal $h^{-1}$.*

At the optimum feed rate, the conversion is 52.9% and the outlet temperature is 157 °F. The occurrence of a maximum molar flow rate of D is expected qualitatively (see Example 5.5.2 in REB, The Book). Reactions (1) and (2) are series-parallel reactions, and D is the intermediate product. At low reaction times (high feed rates), the concentrations of A and B are high leading to a high rate of production of D by reaction (1), and because the concentration of D is small, it's consumption rate is low. As the reaction time increases (feed rate decreases), the rate of generation of D by reaction (1) decreases due to lower concentration of A, while the rate of consumption of D by reaction (2) increases due to higher concentration of D. Eventually, the rate of production of D by reaction (1) decreases to the point where it equals the rate of consumption of D by reaction (2). This is the point where the molar flow of D is maximized. At reaction times higher than this (lower feed rates), the rate of consumption of D is greater than it's rate of generation, so its concentration decreases.

Both reactions are endothermic, but the outlet temperature is significantly greater than the feed temperature. This means that the steam is providing more heat than is being consumed by the reaction. The added heat is likely essential for having a reaction rate that is large enough to get high conversion.

### *Summary*

At a feed rate of 44.8 gal $min^{-1}$, 52.9% of the reactant A is converted. The final temperature is 157 °F.